STEM Exploration for High Schoolers

# Chapter 4

## Worksheet

**Instructions:**

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

**Chapter 4: Engineering Principles - Interactive Worksheet**

Hey, future engineers! Welcome to your mission log for Chapter 4: Engineering Principles. This worksheet is your blueprint to becoming a problem-solving, world-building mastermind. Packed with mind-bending challenges, hands-on tasks, and creative quests, it's designed to make you think like an engineer while having a blast. Grab a pencil, rally your crew (or go solo), and let's construct some epic ideas. From bridges to robots, your journey to innovation starts now!

**Part 1: Engineering Explorer (What is Engineering? Types of Engineering Disciplines)**

**Mission**: Uncover the world of engineering and find your superpower!

- **Brain Blast (3 mins)**: List 3 types of engineering disciplines from the chapter. Pick one and write how it could help solve a teen problem (like slow Wi-Fi or boring commutes).
  - Type 1: ____________________
  - Type 2: ____________________
  - Type 3: ____________________
  - My Pick & Solution:
- **Real-World Recon**: Think of 2 things in your life (like your phone or a bridge near you) that engineers created. Which type of engineer made each? Ask a friend or family member if they know—write their guess!
  - Thing 1: ____________________ (Engineer Type: ____________________) (Their Guess: ______________________________________________________)
  - Thing 2: ____________________ (Engineer Type: ____________________) (Their Guess: ______________________________________________________)
- **Dream Job Debate**: If you could be any kind of engineer, which would you pick and why? Write 2 sentences (e.g., "I'd be a software engineer to design games!").
  - My Choice:

**Part 2: Design Detective (Design Thinking: Problem Identification and Solution Development)**

**Mission**: Solve problems like a pro with Design Thinking!

- **Problem Spotter**: Think of 1 annoying issue at school or home (like messy lockers or no good study spots). Write a clear "problem statement" as described in the chapter (e.g., "Teens need a way to organize lockers without losing stuff.").
  - My Problem Statement:
- **Idea Storm (5 mins)**: Brainstorm 3 wild solutions for your problem—no limits, even if they're silly (like a

robot locker buddy). Circle your fave!

  - Idea 1: ____________________
  - Idea 2: ____________________
  - Idea 3: ____________________

- **Empathy Check**: Talk to 1 person (friend, sibling, parent) about your problem. What's 1 thing they said bugs them about it? How could your fave idea help them?
  - Their Comment:
  - How My Idea Helps:

**Part 3: Structure Sleuth (Basics of Structural Engineering)**

**Mission**: Crack the code of building strong stuff!

- **Force Finder**: Look at a chair, table, or bridge near you (or picture one). What forces (like gravity or weight) does it face? How does its shape or material fight those forces? Write 2 observations.
  - Observation 1:
  - Observation 2:
- **Triangle Trick**: Draw a quick sketch of a bridge or tower with triangles for stability (like a truss design). Label where you think it's strongest. Why did you put triangles there?
  - (Sketch space below or on separate paper)
  - Reason:
- **Mini Challenge**: Using 10 straws or sticks and some tape, build a tiny tower (6 inches tall max). Test it with a small weight (like a coin). Did it hold? If it fell, what would you fix?
  - Result:
  - Fix Idea:

**Part 4: Robot Ranger (Introduction to Robotics)**

**Mission**: Gear up to build machines that move and groove!

- **Robot Anatomy**: Name 3 key robot parts from the chapter. Pick 1 part and explain how it's like something in a human (e.g., "Sensors are like eyes.").
  - Part 1: ____________________
  - Part 2: ____________________
  - Part 3: ____________________
  - Comparison:
- **Bot Brainstorm**: Imagine a robot for teens (like a homework helper). What's its name, and what's 1 cool thing it does with its sensors or actuators? Draw it if you're feeling artsy!
  - Robot Name: ____________________
  - Cool Feature:
- **Tech Hunt**: Spot 1 robot-like device in your life (like a vacuum bot or smart speaker). What's 1 component it has (like a sensor) and how does it use it?
  - Device: ____________________
  - Component & Use:

**Part 5: Builder's Battle (Hands-On Activity: Designing and Testing a Small Bridge or Robot)**

**Mission**: Create and conquer with a real engineering project! Choose 1 below (or both for extra cred). **Option 1: Bridge Builder**

- **Task**: Build a mini bridge (12 inches span) with straws, popsicle sticks, tape, etc., to hold 1 pound (like small books). Test between 2 surfaces (like books).
- **Quick Plan**: What design (beam, arch, truss)? Sketch it quick!
  - Design Type: ____________________
  - (Sketch space below or on separate paper)
- **Test Log**: How much weight did it hold? What broke first?
  - Weight Held: ____________________
  - What Broke:
- **Upgrade**: How would you make it stronger? (e.g., "Add more triangles!")
  - Idea:

**Option 2: Robot Rookie (Brush Bot)**

- **Task**: Make a tiny bot with a toothbrush head, small motor, and coin battery (get adult help if needed). Test if it moves on a table.
- **Quick Plan**: Where will motor/battery go? Sketch it quick!
  - (Sketch space below or on separate paper)
- **Test Log**: Did it move? How far in 10 seconds? Any flips or fails?
  - Movement: ____________________
  - Issues:
- **Upgrade**: How could it be cooler or better? (e.g., "Add weight for speed!")
  - Idea:

**Show-Off Bonus**: Name your creation (bridge or bot) and snap a pic (if allowed) to share with class or friends. What's its epic name?

- Name: ____________________

**Part 6: Future Forge (Engineering Tomorrow)**

**Mission**: Predict and invent the future of engineering!

- **Trend Tracker**: Pick 1 future engineering idea (like self-building houses or medical robots). How could it change your life in 10 years?
  - Idea: ____________________
  - Change:
- **Crazy Combo**: Mash up 2 random things (like "sneakers + robotics") into a futuristic gadget. What does it do?
  - Mashup: ____________________
  - What It Does:
- **Ethics Clash**: If engineers make a robot teacher that knows everything, should it replace human teachers? Write 1 pro and 1 con.
  - Pro:
  - Con:

**Bonus Round: Engineer's Epic Badge**

**Mission**: Claim your badge by reflecting on your engineering adventure!

- **Mind-Blown Moment**: What’s 1 thing from this chapter that amazed you about engineering? (e.g., “I didn’t know robots could be so simple to build!”)
    - My Wow:
- **Next Build**: What’s 1 engineering skill or project you want to try next (like joining a robotics club or designing an app)? How will you start?
    - My Goal:
    - First Step:

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